



Physicochemical characteristics of the active fractions of polyphenols from Arctic macrophytes

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Abstract

Brown algae of the Fucaceae family, collected in the western segment of the Arctic, contain a large number of polyphenolic compounds—phlorotannins—with a unique chemical nature, macromolecular structure, and physicochemical properties, exhibiting bioactive properties. The peculiarity of the biopolymers' properties and structure, which are an element of fundamental scientific research of the cycle, structure—functional nature—properties, determines the priority of the directions of use of polyphenols as effective biologically active pharmaceutical preparations. This work aims to develop methods for selective extraction of phlorotannins from the Arctic brown alga *Fucus vesiculosus*, characterization of physicochemical properties, and biological activity of these compounds. The efficiency of solid-phase extraction (SPE) on a hydrophobic sorbent with grafted hexadecyl groups was shown for fractionation of the polyphenolic extract: active fractions were obtained, characterized by a high content of polyphenols (yield of polyphenols up to 98% relative to the content of these components in algae) with mass-average molecular weights of 21.4–35.8 kDa and oligomeric components in the mass range 374–994 Da. The IR spectra of the fractions were characterized, and the tendency of phlorotannins to form aggregates at high concentrations of the extract was shown. During hydrodynamic studies, a strong polyelectrolyte swelling of polyphenols (PP) was noted. The obtained polyphenolic fractions differ in physicochemical characteristics (acid-base and redox properties) and antioxidant activity.

Keywords Phaeophyceae · Polyphenols · Phlorotannins · Solid-phase extraction · Physicochemical properties · HPLC-MS · Redox potential · Acid-base properties · Antioxidant activity

Introduction

Brown algae are a unique raw material for a variety of substances with a wide range of consumer properties. The component composition of Arctic brown algae includes polyphenolic compounds—phlorotannins (polymers of phloroglucinol)—with content in the biomass reaching 20% (Ragan and Glombitza 1986). According to the structure and polymer properties, phlorotannins represent an extensive group of molecules

that differ in the degree of polymerization, the nature of the bonds between the phloroglucinol units, and the number of hydroxyl groups (Padua et al. 2015). The molecular weight of brown algal polyphenols varies in a wide range from 126 Da to 650 kDa (Boettcher and Targett 1993). Assessment of the chemical profiles variability of low molecular weight brown algal phlorotannins during the year showed no significant fluctuations in the oligomeric composition of algal polyphenols (Kirke et al. 2019).

Despite extensive studies in the field of phlorotannins, their physicochemical properties and relationship with bioactivity are still poorly understood. Ongoing research in this area of phenolic compounds has focused on polyphenols derived from terrestrial plants such as flavonoids and phenolic acids (Masek et al. 2011; Zhang et al. 2011).

At the same time, structural differences in phlorotannin macromolecules determine the variety of biological activity of these compounds, including strong antioxidant properties, hepatoprotective, anti-allergic, anti-tumor, anti-inflammatory, anti-diabetic, anti-fungal, and other activities (Shibata et al.

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2008; Lee et al. 2010; Bae 2011; Gupta and Abu-Ghannam 2011; Kim and Himaya 2011; Eom et al. 2013; Lee and Jeon 2013; Zenthoefer et al. 2017; Abdelhamid et al. 2018; Barbosa et al. 2018; Lee et al. 2018; You et al. 2018; Liu et al. 2019). The use of phlorotannins as antibacterial drugs has been shown to be promising (Bogolitsyn et al. 2019a). It is important to note that phlorotannins have no cytotoxicity and treated cells remain metabolically active (Kim et al. 2018; Lee et al. 2018; Bogolitsyn et al. 2019a). The antioxidant properties of phlorotannins are related mainly to their antiradical effect, which is manifested in the protection of the body from damage by reactive oxygen species.

The relationship between the molecular weight and antioxidant activity of brown algal phlorotannins is still under study (Wang et al. 2012). Thus, it was noted a higher antioxidant activity of high molecular weight phlorotannins compared to the low molecular weight polyphenolic fraction (Ferrerres et al. 2012; Kirke et al. 2019). However, in other studies, it was shown that an increase in the molecular weight of phlorotannins leads to a decrease in antioxidant ability (Nakamura et al. 1996; Shibata et al. 2008; Liu and Gu 2012; Abdelhamid et al. 2018). When conducting such studies, Wang et al. (2012) did not find a clear relationship between the antioxidant ability and the molecular weight of phlorotannins.

Analysis of the biological activity of polyphenols suggests that it is due to features of acid-base and redox properties of phlorotannins, as well as the formation of a macromolecular structure in the process of self-organization of a biopolymer monomer units with the involvement of phenolic OH-groups in the formation of hydrogen bonds in the macromolecule.

Therefore the goal of this work is the development of methods for selective extraction of active phlorotannin fractions from Arctic brown algae *Fucus vesiculosus*, characterization of physicochemical properties, and biological activity of these compounds.

Experimental

Materials and reagents

Air-dried *Fucus vesiculosus* were used. Macrophytes were selected during the Arctic Floating University integrated research expeditions in 2017 from the waters of the Bolshoi Solovetsky Island of the White Sea (Russian Federation) (65°02' N; 35°69' E).

Reagents *N,N*-dimethylformamide (HPLC grade), chloroform (> 99%), acetone (> 99%), isopropanol (> 99%), and ethyl acetate (> 99%) were from Component-Reactive (Russia). Acetonitrile (HPLC grade) was from NPK Kriochrome (Russia). High purity deionized water was obtained using a Simplicity UV system (Millipore, France).

Formic acid (HPLC grade), Folin-Chocalteu reagent, lithium bromide (> 99%), L (+) - ascorbic acid (99%), 2,2-diphenyl-1-picrylhydrazyl (DPPH), and phloroglucinol (> 99%) were from Sigma-Aldrich (USA). Acetic acid (> 99.8%), hydrochloric acid (35–38%), sodium carbonate (99.8%), sodium hydroxide (98%), and sodium chloride (98%) were from Neva-Reactive (Russia). Ethanol (technical grade, 96.2%) was from the Kirov Biochemical Plant (Russia). Methanol (> 99%) was from Himmed (Russia).

Phlorotannin extraction and fractionation

Algal polyphenol extract was obtained according to the scheme shown in Fig. 1. Air-dried algae were crushed and sieved to a particle size of 0.03–0.20 mm and extracted with chloroform in a Soxhlet for 8 h to remove the lipid-pigment complex. Extracted algae was dried in an oven at 40 °C and ground to a fraction of less than 0.01 mm. Then, the dried and ground algae was extracted with acetone at 50 °C for 1 h with the ratio of mass:volume as 1:20 with constant stirring. The extraction was carried out using a heating magnetic stirrer with a temperature sensor C-MAG HS7 (Ika Werke GmbH, Germany). The extract was decanted into a chemical flask, and the acetone extraction procedure was repeated to maximize the extraction of polyphenols from algae. The obtained acetone extracts were combined and centrifuged at a speed of 9509 *xg* for 15 min using a ProteomLab XL-A centrifuge (Beckman, USA). The purified acetone extract was concentrated using a rotary evaporator IR-1-LT (Labtex, Russia) to a thick state; process parameters—water bath temperature 45 °C, vacuum 0.1 MPa, and rotation speed 75 rpm. Distilled water was added to the thick extract (the mass of water is equal to the mass of the original dry algae) to obtain an aqueous polyphenolic extract. The aqueous polyphenol solution was lyophilized to give a brown powder of the polyphenol complex.

Fractionation of polyphenols was performed by solid-phase extraction. In the course of the work, the following solid phase extraction cartridges were used in parallel with each

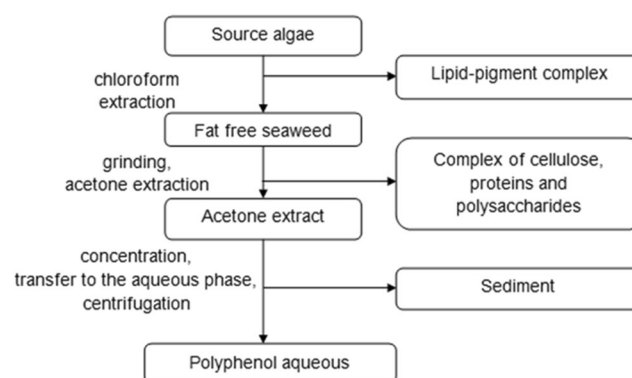


Fig. 1 Scheme for the isolation of polyphenols from brown algae

other to compare fractionation efficiency: Diapak P (BioChemMac, Russia) and Diapak C16 (BioChemMac, Russia). The characteristics of the selected sorbents are presented in Table 1. Before work, the cartridges were preconditioned with 5 mL of methanol and 10 mL of 0.5% aqueous hydrochloric acid solution (pH 2) at flow rate 1–2 mL min⁻¹. To fractionate the aqueous extract polyphenols (fraction 1), 5 mL of a solution of polyphenol extract with a concentration of 1 mg mL⁻¹ was passed through the cartridge, the non-adsorbed fraction (fraction 2) was taken and, if necessary, lyophilized.

After that, the cartridge was washed with 10 mL of distilled water, and the polyphenols (fraction 3) were desorbed with 2 mL of an organic solvent. To identify the optimal organic solvent for the most efficient extraction of polyphenols from the cartridge, methanol, ethanol, isopropanol, acetone, and ethyl acetate were used. Each type of cartridge was tested with each organic solvent. The experiment was carried out in triplicate. Fraction 3 was selected and, if necessary, freeze-dried. The indelible components held on the cartridge are an irreversibly adsorbed fraction (fraction 4).

Assessment of the effect of the pH of the aqueous polyphenol extract (fraction 1) on the sorption efficiency on cartridges showed that the addition of hydrochloric acid into the aqueous extract to pH 2 increases the proportion of sorbed polyphenols to 99% relative abundance. Six experiments were conducted in total; for each cartridge type an experiment was carried out without the addition of acid, at pH 2 and pH 1. Each experiment was carried out in triplicate.

Further studies on the selection of an effective eluent for desorption of polyphenols (separation of fraction 3) made it possible to establish that methanol is the optimal eluent for the studied cartridges. Ten experiments were conducted in total at the optimum pH; for each cartridge type, an experiment was carried out with elution with 5 organic solvents. Each experiment was carried out in triplicate.

Phlorotannin characterization

The total content of the polyphenols in fractions was determined colorimetrically using the Folin–Ciocalteu reagent according to the method of Wang et al. (2012) with modifications. An aliquot of the sample (1 mL, concentration 0.1–

0.8 mg mL⁻¹) was mixed with 5 mL of Folin–Ciocalteu reagent (10% in distilled water). After 5 min, 4 mL of sodium carbonate (7.5% in distilled water) was added. To prepare a blank sample, 1 mL of distilled water was mixed with 5 mL of Folin–Ciocalteu reagent, and after 5 min, 4 mL of sodium carbonate was added. The samples were incubated for 1 h at room temperature in the dark. The absorbance was measured at 765 nm with a Spekol 1300 UV–vis spectrophotometer (Analytik Jena, Germany). The experiment was carried out in triplicate. The calibration range was 20–100 µg mL⁻¹.

The hydrodynamic parameters of the fraction 1 were determined by viscometry, translational diffusion, and high-speed sedimentation:

- For the viscosity measurements, the prepared polyphenol solution was kept for 1 day at room temperature. The intrinsic viscosity $[\eta]$ was determined using an Ostwald capillary viscometer at a temperature of 21 °C in a 3:1 ethanol–water mixture. Solvent density $\rho_0 = 0.856 \text{ g cm}^{-3}$, solvent viscosity $\eta_0 = 1.95 \text{ cP}$.
- The translational diffusion coefficient D was determined by the dispersion of the solution–solvent interface on a Tsvetkov polarizing diffuser at 24 °C. The interface solution–solvent was formed in a 3-cm optical cuvette along the light beam. Photographs of the interference bands of the solution–solvent interface evaluated by the maximum ordinate method and the area under the interference band (Eskin and Frenkel 1979; Tsvetkov 1989).

The hydrodynamic radius of the equivalent sphere R_h was calculated according to the Einstein–Stokes equation:

$$R_h = \frac{kT}{6\pi D\eta_0},$$

where k is the Boltzmann constant, D is the translational diffusion coefficient, η_0 is the viscosity of the solvent, and T is the temperature.

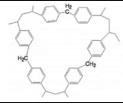
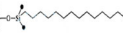
- The sedimentation rate coefficient s of macromolecules in a centrifugal field was measured on a MOM 3180 ultracentrifuge (MOM, Hungary) with a polarization–interferometric attachment at a rotor speed of 40,000 rpm, 24 °C (Eskin and Frenkel 1979; Tsvetkov 1989).

The Svedberg method (Svedberg and Pedersen 1940) determines the absolute molecular weights:

$$M_{sD} = \frac{s}{D} \frac{N_A k T}{1 - v\rho_0},$$

where s is the sedimentation coefficient, D is the translational diffusion coefficient, N_A is the Avogadro number, k is the Boltzmann constant, T is the absolute temperature, and ρ_0 is

Table 1 Characterization of cartridges for TFE

Cartridge	Sorbent	Functional group	Particle size, µm	Pore Size, Å	Surface area, m ² g ⁻¹
Diapak P	hydrophobic sorbent based on hypercrosslinked polystyrene		50–100	10–1000	1132
Diapak C16	hydrophobic sorbent with grafted hexadecyl groups		63–200	100	300

the density of the solvent.

The experiments were carried out in triplicate.

IR spectroscopy: Freeze-dried polyphenol fractions were characterized by IR spectroscopy using an IR-Fourier-spectrophotometer IRPrestige 21 (Shimadzu, Japan) and a single-reflection horizontal attenuated total reflection attachment MIRacle with a ZnSe prism (Pike, USA). IR spectra were recorded at a resolution of 4 cm^{-1} in the range $4000\text{--}600\text{ cm}^{-1}$ with several scans of 128.

The molecular weight distribution of phlorotannins was determined using a size-exclusion chromatography system (Staier, Russia), including a DG18 degasser, a HPLC pump, a Rheodyne 7725i injector, a TS10 column oven, and a spectrophotometric detector (UVV-104.1M). Separations were carried out with two Phenogel columns ($300 \times 7.8\text{ mm}$, $10\text{ }\mu\text{m}$, Phenomenex, USA). The mobile phase was *N,N*-dimethylformamide with lithium bromide (0.1 M LiBr) and 1.0% v/v acetic acid. The mobile phase flow was 0.40 mL min^{-1} , the oven temperature $60\text{ }^{\circ}\text{C}$, the injection volume $100\text{ }\mu\text{L}$, and the detection wavelength of 275 nm . Polystyrene standards of 115 kDa , 42.3 kDa , 18.2 kDa , 9.0 kDa , 6.5 kDa , 2.8 kDa , and 940 Da were used for calibration. Freeze-dried sample mass 1 mg was dissolved in 1 mL of mobile phase, filtered through a $0.22\text{-}\mu\text{m}$ nylon syringe filter, and injected into the chromatographic system. Data acquisition and calculations were carried out using Multichrome software (Ampersand, Russia). The experiment was carried out in triplicate.

Particle size: The particle size of phlorotannins was determined by dynamic light scattering (quantum correlation spectroscopy) using a Horiba LB-550 laser analyzer (Horiba, Japan), which evaluates both the particle size distribution in the diameter range $D = 1\text{--}6000\text{ nm}$ and their contribution to light scattering. Particle size was estimated in aqueous solutions with a concentration of the determined components of 1 g L^{-1} (high concentration) and $50\text{--}100\text{ mg L}^{-1}$ (low concentration).

HPLC conditions: Chromatographic analysis was performed using a LC-MS system consisting of a LC-20 “Prominence” liquid chromatograph (Shimadzu, Japan) equipped with an SPD-M20A diode array spectrophotometric detector and a mono-quadrupole mass spectrometer LCMS-2020 (Shimadzu, Japan). The separation was carried out with a Zorbax Bonus-RP column ($150 \times 300\text{ mm}$) with a particle size of $3.5\text{ }\mu\text{m}$ (Agilent, USA). The mobile phase consisted of deionized water containing 0.1% vol. formic acid (A) and acetonitrile (B). The gradient elution program was $0\text{--}6\text{ min}$, 10% B; $10\text{--}15\text{ min}$, 30% B; and $20\text{--}25\text{ min}$, 60% B. Detection was done using the $200\text{--}400\text{ nm}$ wavelength range and the analytical wavelength was 280 nm . The conditions of the mass-spectroscopic detector were electrospray ionization in the negative mode -4.5 kV ; desolvation line, $350\text{ }^{\circ}\text{C}$; and heating block, $350\text{ }^{\circ}\text{C}$.

Acid-base properties: The pK_a values of phenolic hydroxyl groups in macromolecules were determined by spectrophotometric titration. The dissociation constants of polyphenols were determined as follows: an aqueous solution of the compounds with a concentration of 0.3 mg L^{-1} was placed in a cell (100 mL) kept at $25\text{ }^{\circ}\text{C}$. The experiment was carried out with constant nitrogen sparging. Direct titration was performed with sodium hydroxide at a titrant concentration in the range of $10^{-3}\text{--}4\text{ M}$. A pH-211 ionomer (Hanna, Romania) with a combined glass electrode was used to measure pH. After adding each portion of the titrant, the pH of the solution was measured, 3 mL were withdrawn, and the spectrum (relative to water) was recorded on a UV 1800 spectrophotometer (Shimadzu, Japan) at wavelengths from $220\text{ to }500\text{ nm}$. pK_a values were calculated by regression analysis. The experiment was carried out in triplicate. The standard deviation of pK_a values does not exceed 0.2 units.

$$A = A_{\kappa} + \sum_{i=1}^n \left(\frac{\Delta A_i}{1 + 10^{pK_{ai} - pH}} \right),$$

where ΔA_i is the change in optical density caused by the absorption of the corresponding phenolic structure of polyphenols and ΔA_{κ} is the optical density of the non-ionized form of polyphenols.

The relative content of phenolic hydroxyl groups which had uninvolved and involved in intramolecular H-bonds was determined by the ratio of the corresponding changes in the values of A_i .

Redox potential: The effective oxidation/reduction potential (ORP) of polyphenols was determined by indirect oximetry in the potassium ferricyanide-potassium ferrocyanide system ($\text{K}_3[\text{Fe}(\text{CN})_6]$ - $\text{K}_4[\text{Fe}(\text{CN})_6]$) with a ratio of $10^{-3}/10^{-5}\text{ M}$ in 0.01 M KOH at $25\text{ }^{\circ}\text{C}$ with constant bubbling (passing through a solution) with nitrogen to displace air oxygen to prevent oxidation of dissolved polyphenols. The Expert-001.1.01 ionomer (EconixExpert, Russia) and the electrode system (a platinum electrode and an auxiliary silver-silver electrode) were used to measure the potential of the redox system. The effective ORP was calculated using pseudo-equilibrium between the redox system, the investigated substrate, and the initial products of its oxidation (Belyi et al. 2010; Bogolitsyn et al. 2014). The experiment was carried out in triplicate. The standard deviation of the values of the effective potential φ_{02}^* does not exceed 5 mV .

$$\begin{aligned} \varphi_{02}^* = & \varphi^{init} - \frac{RT}{F} \ln \frac{c_{03}^{init}}{c_{\Sigma} - c_{03}^{equil}} + \frac{RT}{F} \ln \frac{c_{03}^{equil}}{c_{\Sigma} - c_{03}^{equil}} \\ & + \frac{RT}{2F} \ln \frac{2c_{ArO}^{init} - 4c_{03}^{init} + 4c_{03}^{equil}}{c_{03}^{init} - c_{03}^{equil}} + \frac{2,303RT}{2F} pH, \end{aligned}$$

where φ^{init} is the initial ORP of the potassium ferricyanide-ferrocyanide system, mV ; c_{03} , c_{04} , c_{Σ} is analytical

concentrations of oxidized, reduced forms and their sum of the mediator system, mol L^{-1} ; c_{ArO}^{init} is the initial phenol concentration, mol L^{-1} ; R is the universal gas constant, $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$; and F is the Faraday number, $F = 96.485 \text{ C mol}^{-1}$.

Antioxidant activity. The radical scavenging (antioxidant) activity was determined using a spectrophotometric method by the degree of decolorization of the DPPH solution, according to Wang et al. (2012), with modifications. Briefly, 0.1 mL of the sample with a concentration of 0.1 mg mL^{-1} was mixed with 3.9 mL of the DPPH solution ($15 \text{ } \mu\text{M}$ in methanol, prepared daily). After 30 min at room temperature in the dark, the absorbance was measured at 515 nm with a Spekol 1300 UV-vis spectrophotometer (Analytik Jena, Germany). The experiment was carried out in triplicate. Ascorbic acid was used as the reference standard. The results are expressed as mg ascorbic acid per 1 g of dry extract.

Results

The scheme for isolating polyphenolic compounds from *F. vesiculosus* (Fig. 1) allows the selective extraction of polyphenols from macrophyte biomass with a fraction of the target component $84.0 \pm 1.9\%$ in the aqueous polyphenol extract (fraction 1).

Characterization of the hydrodynamic and conformational properties of polyphenol macromolecules in aqueous solutions

The structure of phlorotannins in fraction 1 was studied using hydrodynamic methods—viscometry, translational diffusion, and sedimentation velocity. At the initial concentration of 0.789 g dL^{-1} , the solution was molecular but opaque, without delamination or precipitation. Further dilution increased transparency, while the flow time of the solution remained constant and was reproduced with repeated measurements for 7 days. Figure 2 shows a graph of the changes in viscosity of fraction 1.

The suppression of electrostatic interactions was carried out by adding a low molecular weight electrolyte—NaCl—which increases the ionic strength of the solution at low polymer concentrations. However, the presence of salt did not cause a significant reduction in the size of the molecules. The value $[\eta] = 0.1 \text{ dL g}^{-1}$ was taken as an estimated value.

To estimate molecular weight, according to Svedberg (Svedberg and Pedersen 1940), it is necessary to know the polymer buoyancy factor ($1 - v\rho_0$) or its density. Structurally similar polyphenylenes (3 and 4 benzene rings) and polystyrene (8 benzene rings) were used to determine the buoyancy factor. Thus, it is possible to postulate a range of calculated molecular weights. The polyphenylene sample (4 benzene

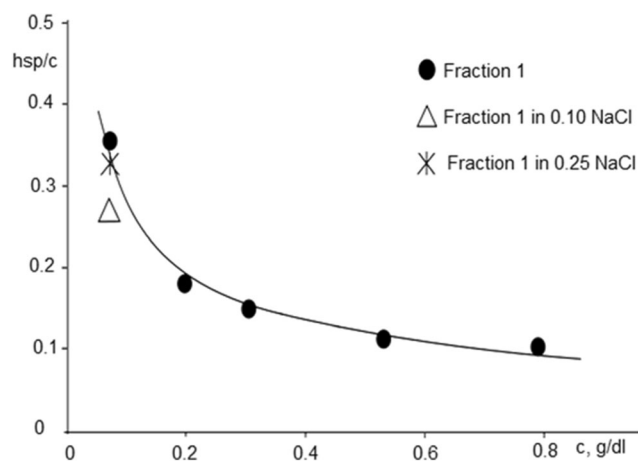


Fig. 2 Concentration dependence of the reduced viscosity of polyphenols

rings) with a buoyancy factor of 0.366 and a molecular weight of 24 kDa is the closest in structure to the polyphenols under study.

The magnitude of the hydrodynamic invariant indicates that the measurements of sedimentation diffusion and viscosity, determined in independent experiments, are consistent. The best correlation of hydrodynamic characteristics at $A_0 = 3.7 \cdot 10^{10} \text{ erg K}^{-1}$, which corresponds to a molecular mass of 24 kDa (Table 2). However, it is worth noting that all values were obtained with unsuppressed polyelectrolyte effects.

The results of comparing the efficiency of fractionation of an aqueous polyphenolic extract using solid-phase extraction cartridges Diapak C16 and Diapak-P are presented in Table 3. With a hydrophobic sorbent based on the hypercrosslinked polystyrene (Diapak P), up to 100% relative abundance of polyphenols are sorbed from fraction 1, so fraction 2 is absent. After desorption, up to 60% relative abundance of polyphenols (fraction 3) is released from the cartridge but up to 45% relative abundance of polyphenols is irreversibly sorbed (fraction 4), which does not allow repeated use of this sorbent.

A hydrophobic sorbent with grafted hexadecyl groups (Diapak C16) allows the collection of the non-adsorbable material (fraction 2, up to 57% relative abundance) and the absorbed fraction (fraction 3, up to 25% relative abundance), while only 18% relative abundance of polyphenols are irreversibly sorbed (fraction 4). Both recoverable fractions 2 and 3 (Table 3) are characterized by a high degree of purity (up to 98%).

The results obtained by evaluating the effectiveness of the separation of polyphenols from an aqueous extract by the SPE method suggest that Diapak C16 cartridges are optimal. Further studies were aimed at studying the physicochemical characteristics of the aqueous polyphenolic extract (fraction 1) and fractions 2 and 3 isolated using Diapak C16 cartridges.

Polyphenol fractions 1, 2 and 3 were characterized by IR spectroscopy. The IR spectra are similar for both fractions and have the same characteristic absorption bands. The spectrum

Table 2 Hydrodynamic characteristics of the polyphenolic fraction 1

$[\eta]$, dL g ⁻¹	$D \cdot 10^7$, cm ² s ⁻¹	R_h , nm	dn/dc , cm ³ g ⁻¹	s, Sv	M_{SD}	$A_0 \cdot 10^{10}$, erg K ⁻¹
0.1	4.26	2.6	0.148	1.5	40,000	4.4
					30,000	4.0
					24,000	3.7

of fraction 3 is shown in Fig. 3; several main characteristic bands are identified, and their interpretation is presented in Table 4.

The molecular weight distribution profiles are almost identical for fractions 1 and 2 (Fig. 4). The weight average molecular weight for the largest peak (Table 5) of the non-adsorbable fraction 2 is 53.8 kDa. The molecular weight distribution for fraction 3 is different, peaks 1 and 2 are practically absent, and the mass average molecular weight for the largest peak is 11.7 kDa.

The polyphenolic fractions tested by dynamic light scattering indicated that fractions 2 and 3 behave similarly (Table 5). In concentrated aqueous solutions (1 g L⁻¹), the particle size range was from 200 nm to 6 μm or more. Larger particles made a more significant contribution to light scattering due to their size, although particles with a size of 200–1000 nm were the most abundant. In aqueous solutions with a concentration of 50–100 mg L⁻¹, a group of particles with a size in the range of 6–16 nm predominates (Fig. 5).

The component composition of fractions 1–3 was analyzed using liquid chromatography-mass spectrometry. Oligomeric phlorotannins of 374–994 Da were found in these fractions; the chromatograms of fraction 3 are shown in Fig. 6.

Assuming similar extinction coefficients of polyphenolic compounds in this mass range and the similarity of their structures, the ratios of oligomeric phlorotannins in the fractions 1–3 were determined using the areas of chromatographic peaks obtained with a diode-matrix photometric detector (Fig. 7).

The acid-base characteristics of polyphenolic fractions showed that all of them had differences in dissociation constants, with the smallest values in adsorbed fraction 3 (Table 5). At the same time, fraction 3 has the highest content of free OH-groups (44%).

According to the data in Table 5, all polyphenol fractions have a high redox potential of 868–879 mV, characteristic for high molecular weight compounds. The differences in the redox potentials are insignificant and are within the error range, although there is a tendency for higher redox potential with increasing molecular weight (fractions 2 and 3).

Characterization of the antioxidant properties (Table 5) of studied polyphenolic fractions revealed a high activity of fraction 3 (862 ± 34 mg of ascorbic acid per 1 g of extract), which has a lower weight average molecular weight (21.4 kDa).

Discussion

As far as the author is aware, there is very little literature on the study of the physicochemical properties of brown algal polyphenols. A comprehensive study of this fundamental area is important to identify the properties and structure of phlorotannin macromolecules, which are an element of scientific research of the structure—functional nature—properties cycle and determine the priority of using brown algal polyphenols as effective biologically active ingredients for the pharmaceutical industry.

Table 3 Evaluation of the effectiveness of Diapak C16 and Diapak P cartridges for TFE polyphenols from an aqueous extract of polyphenols

Fraction	PP content		The purity of the fraction, % PP from the dry extract
	% of the content in fraction 1	% of the content of adsorbed PP	
Fraction 1	100.0	—	84.0 ± 1.9
Diapak P			
Fraction 2	1.1 ± 0.3	—	—
Fraction 3	58.1 ± 2.4	58.7 ± 2.4	94.9 ± 2.6
Fraction 4	40.8 ± 2.1	41.3 ± 2.1	—
Diapak C16			
Fraction 2	56.9 ± 2.9	—	96.2 ± 2.9
Fraction 3	25.5 ± 1.1	59.2 ± 2.6	98.0 ± 1.8
Fraction 4	17.6 ± 0.9	40.8 ± 2.1	—

±—Std deviation, $n = 3$

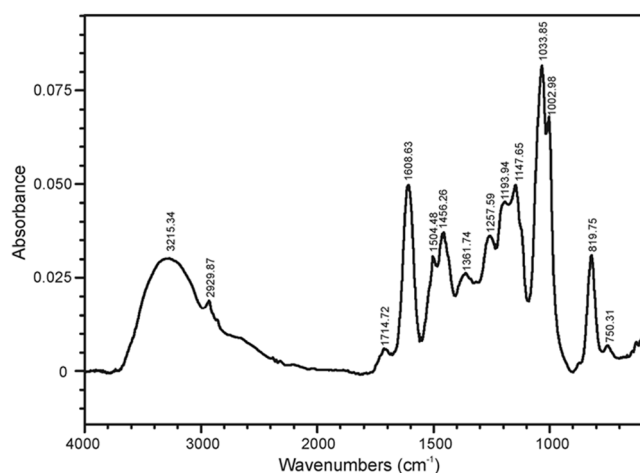


Fig. 3 IR spectrum of fraction 3

The conformational properties of molecules are manifested in translational and rotational friction; therefore, molecular hydrodynamics methods—sedimentation velocity, isothermal translational diffusion, and capillary viscometry—are used to study these features. These methods and modern theories of hydrodynamics were used to estimate the size and conformation of macromolecular tangles crucial for establishing the topological structure of polymers. Hydrodynamics methods remain the experimental basis for the majority of studies regarding the properties of biopolymers.

Determination of the hydrodynamic characteristics of the polyphenolic extract enables the analysis of the phlorotannin macromolecule structure. The graph of the concentration dependence of the reduced viscosity of polyphenols (Fig. 2) shows an increase in viscosity with decreasing concentration. This may be associated to an increase in the size of the molecular coil with a decrease in the ionic strength of the solution—the effect of polyelectrolyte swelling. The reason is the presence of charged OH-groups in the molecules. This type of dependence does not allow to assess the intrinsic viscosity and size of the macromolecules correctly. At the same time, the results of the hydrodynamic testing allow to assume that polyphenols have a flexible chain structure, which tends to fold into a macromolecular ball, with a higher density as the

molecular weight of the sample increases and the intramolecular bonds are formed. The mutual screening and binding of active phenolic OH-groups during the formation of a high molecular weight macromolecular coil lead to a decrease in the biological activity of polyphenolic compounds, which is consistent with previous studies (Bogolitsyn et al. 2019a, b). The hypothesis of the low availability of hydroxyl groups in larger macromolecules due to the use of active centers in structural bonds between monomeric units or steric hindrances was also presented in Heffernan et al. (2015) and Kirke et al. (2017).

Currently, the solid-phase extraction method is widely used for the selective isolation of active fractions, which allows to obtain fractions with various physicochemical properties. For successful fractionation of biologically active substances, it is necessary to competently conduct optimization of the conditions of the separation process.

The SPE sorbents used do not have ionizable groups and realize a reverse-phase regime based on hydrophobic physical interactions. The results of the conditions selection showed that acidification of the extract to pH 2 increases the sorption of polyphenols due to the suppression of the dissociation of charged hydroxyl groups of phlorotannins and their transition to a molecular state (Tsao 2010; Vavouraki 2020).

The presence of charged hydroxyl groups reduces the tendency of hydrophobic interactions for polyphenols; therefore, the conversion of phlorotannins into a molecular form increases their tendency for sorption (Mijangos et al. 1997; Ferri et al. 2011; Kaleh and Geißen 2016).

It was found that the degree of sorption of polyphenols on the selected sorbents is directly proportional to their surface area (Diapak C16 < Diapak P). The sorbent Diapak P based on hypercrosslinked polystyrene demonstrates a tendency to hydrophobic and π - π interactions, which, combined with a high surface area, explains the difficult elution of phlorotannins ($60.4 \pm 3.6\%$ is desorbed). Since up to 40% of polyphenols are irreversibly adsorbed on this cartridge, the possibility of reusing this type of sorbent is uncertain.

The conducted research demonstrates that the hydrophobic sorbent with grafted hexadecyl groups and the smallest

Table 4 Interpretation of the characteristic bands of the obtained IR spectra

The position of the strip, cm^{-1}	Assignment of absorption bands
3200–3300	Stretching vibrations of hydroxyl groups involved in hydrogen bonding
2928, 2875–2850	Stretching vibrations of C-H bonds in methyl and methylene groups
1714	Valence vibrations of carbonyl groups, non-conjugated ketones
1606, 1505, 1456	Skeletal vibrations of an aromatic ring
1680	Valence vibrations of O-H bonds of water
1003, 1030, 1150	Valence vibrations of the C-O bond
820, 745	Off-plane deformation vibrations of the C-H bond in the aromatic ring
686, 620	Skeletal vibrations of the C-C bond

Table 5 Complex characteristics of polyphenolic fractions

Characteristic	Fraction		
	1 Polyphenol aqueous extract	2 Nonsorbable fraction	3 Sorbable fraction
Purity, % PP of dry extract	84.0 ± 1.9	96.2 ± 2.9	98.0 ± 1.8
MM weighted average, kDa	32.9 ± 3.3	35.0 ± 3.4	21.4 ± 1.9
MM of the largest peak, kDa	49.8 ± 4.3	53.8 ± 4.9	11.7 ± 1.2
Particle size, nm (concentration 1.0 g L ⁻¹)	u.d.	170–880	225–1000
Particle size, nm (concentration 50 mg L ⁻¹)	u.d.	7.6–17.0	6.6–15.0
pK _{a1} / pK _{a2}	9.5/11.1	9.7/11.7	9.2/11.3
Share free / related OH-groups, %	31 / 69	24 / 76	44 / 56
Redox potential, mV	879 ± 2	875 ± 2	868 ± 2
antioxidant properties, mg of ascorbic acid per 1 g of extract	461 ± 29	694 ± 25	862 ± 34

u.d. – undefined; ± – Std deviation, *n* = 3

surface area allows to isolate two active polyphenol fractions—non-adsorbed and adsorbed—with different characteristics (Table 5, Figs. 4, 6). Retention on this sorbent is carried out mainly due to hydrophobic interactions with hexadecyl groups, which explains the relatively low sorption of phlorotannins.

A study of the particle size of phlorotannins showed that particles ranging in size from 200 nm to 6 µm were observed at high concentrations of the extract. It can be assumed that aggregation occurs with the formation of macromolecular tangles due to the interaction of free OH-groups of individual macromolecules (Kirke et al. 2017). With a decrease in the concentration of the studied extract, aggregate destruction is observed, and a group of particles with a size of 6–16 nm is formed (Table 5, Fig. 5). The results obtained may indicate a high biological activity of phlorotannin molecules even at a low concentration of the extract since functional hydroxyl groups of the molecules are less involved in the formation of intra- and intermolecular hydrogen bonds.

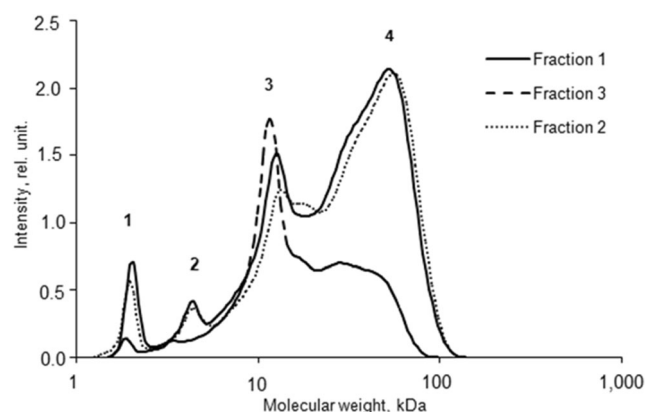
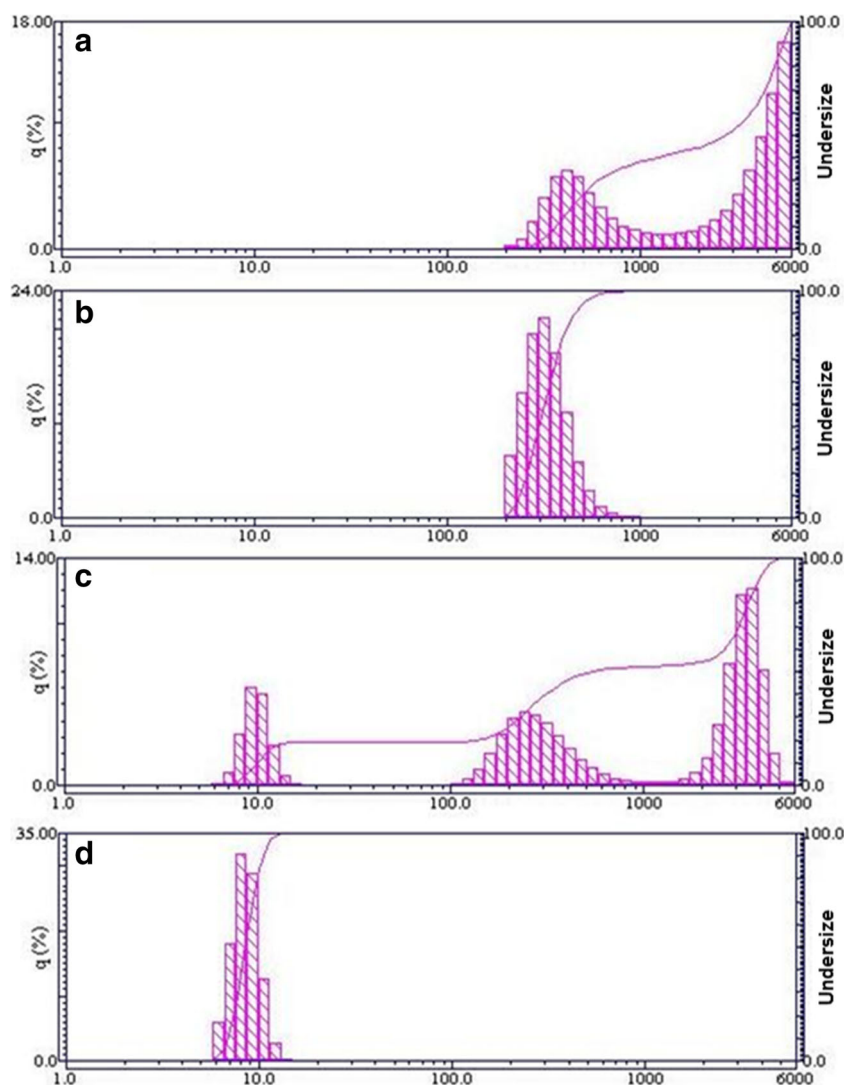


Fig. 4 Comparison of the molecular weight distribution of fractions 1–3 obtained on Diapak C16

The research of the component composition of the fractions revealed that the non-adsorbed fraction 2 is characterized by a low content of oligomeric phlorotannins and a high molecular weight, while fraction 3, on the contrary, is enriched in oligomeric polyphenols and has the smallest molecular weight. The component composition data are consistent with the previous results when the presence of low molecular weight components with 4–8 phloroglucinol rings was noted in the polyphenolic fraction of algae *F. vesiculosus* (Kirke et al. 2019). Similarly, more than 30 phlorotannins in the mass range 247–993 Da and 13 phlorotannins with masses of range 374–870 Da were found in an algal extract of *F. vesiculosus* by HPLC-MS/MS (Catarino et al. 2019). The content of oligomeric components in polyphenolic fractions makes a significant contribution to their activity (Bogolitsyn et al. 2019b); for example, the authors also note a high level of antioxidant activity of fractions enriched in low molecular weight phlorotannins (Kirke et al. 2019).

Brown algal polyphenols are characterized by two dissociation constants pK_{a1} and pK_{a2}, the first referring to free OH-groups and the second to hydroxide groups in a bound state. The study of the acid-base properties of non-adsorbed fraction 2 showed that it had the highest pK_a values and the content of bound OH-groups compared with sorbed fraction 3 (Table 5). Large pK_a values may be due to the electrostatic factors and the involvement of hydroxyl groups in the formation of hydrogen bonds due to the high molecular weight of the fraction (formation of a macromolecular coil) (Perrin et al. 1981). Based on this, it was proposed that phlorotannins with higher dissociation constants will exhibit the least biological activity. The lower molecular weight fraction 3 is most likely to be highly active due to the relatively low pK_a values, which is also evidenced by the large proportion of free hydroxyl groups.

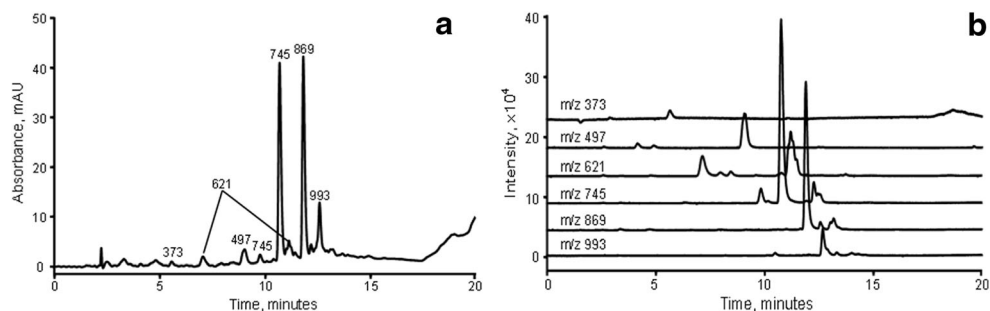
Fig. 5 Dispersed parameters of fraction 3. **(a)** contribution of particles to light scattering, concentration 1 g L^{-1} . **(b)** particle size distribution, concentration of 1 g L^{-1} . **(c)** the contribution of particles to light scattering, a concentration of 50 mg L^{-1} . **(d)** particle size distribution, concentration of 50 mg L^{-1}



The electrochemical analysis is widely used to evaluate the reducing activity of biologically active substances since most types of activities are closely related to redox or electron transfer (Masek et al. 2011; Zhang et al. 2011; Mendoza-Wilson et al. 2014). High ORP values of polyphenolic fractions (Table 5) are characteristic of high molecular weight compounds and can be explained by the conjugation effect. The differences in the values of the redox potentials of polyphenolic fractions are only 4–7 mV; therefore, the influence of

macromolecular factors on the oxidation of phenolate-anions in polyphenols is not pronounced, which is due to several reasons. First, phlorotannins have an ordered conjugated structural system in which the charge transfer rate during delocalization is high; therefore, all phenolate-anions in the polyphenol macromolecule in the ionized state are simultaneously involved in oxidation (Quideau et al. 2011). Secondly, phlorotannins demonstrate high availability of phenolic OH-groups, since there are no carboxyl groups in

Fig. 6 Chromatograms of fraction 3: **(a)** signal of the diode-matrix detector at 220 nm. **(b)** signal of the mass spectrometric detector



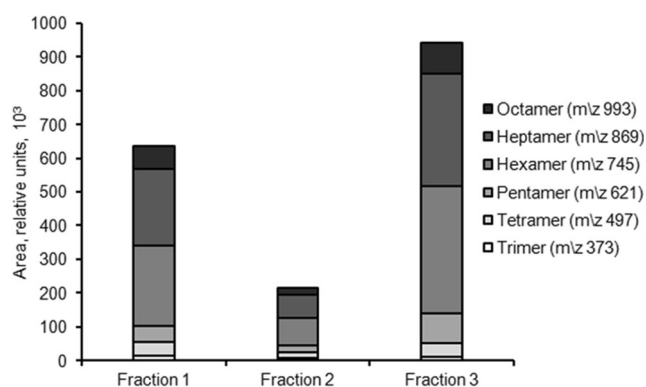


Fig. 7 The ratio of oligomeric phlorotannins in polyphenolic fractions obtained during solid-phase extraction

polyphenols capable of forming stronger intra- and intermolecular hydrogen bonds. The high availability of the functional groups of phlorotannins indicates the supposed high activity of the extracted fractions. A study of the redox processes of some phlorotannins was first conducted relatively recently (Zhang et al. 2018). The authors of this research studied the processes of proton dissociation and electron transfer of phloroglucinol and diacol under various pH conditions and the following conclusion was drawn: the interphloroglucinol bond in the molecular structure of diekol plays an important role in improving the redox activity compared with the phloroglucinol molecule.

To confirm the connection of the physicochemical properties of polyphenols with their biological activity, the antioxidant properties of the fractions were studied (Table 5). Thus, our assumptions were confirmed for the significant activity of the polyphenolic fractions, as well as for higher activity of fraction 3. Fraction 3 is characterized by the highest antioxidant activity, the smallest molecular weight, high content of oligomeric polyphenols, and a greater proportion of free hydroxyl groups, compared with the activity of ascorbic acid, while the higher molecular weight fraction 2 exhibits less antioxidant properties. The results obtained are consistent with the hydrodynamic studies carried out in this work and the previously revealed dependence of antioxidant activity on the molecular mass of phlorotannins when polyphenols with an average molecular weight in the range of 8–18 kDa (monomodal molecular weight distribution) was shown to have the highest biological activity (Bogolitsyn et al. 2019b). A decrease in antioxidant activity with an increase in the molecular weight of brown algal polyphenols was also noted in some works (Nakamura et al. 1996; Shibata et al. 2008; Liu and Gu 2012; Abdelhamid et al. 2018).

The results of this research of the physicochemical properties of phlorotannins and the spatial structure of macromolecules of this component of brown algae are useful for understanding their relationship with the activity of phlorotannins in their biological and chemical processes.

Conclusions

Research conducted in this work showed the possibility of using SPE to obtain active fractions of phlorotannins from a polyphenolic extract of brown algae. Of the studied sorbents, the most effective is the hydrophobic sorbent with grafted hexadecyl groups, which allows to obtain two active fractions of phlorotannins with a high content of the target component.

A comprehensive analysis of the physicochemical characteristics of phlorotannins has been performed using modern analytical methods. The hydrodynamic characteristics of phlorotannin macromolecules and the component composition of the fractions, redox, acid-base and antioxidant properties of polyphenols were studied. The relationship of physicochemical properties with the biological activity of these compounds is shown. The results obtained are new and further detailed studies in the physicochemical properties area of brown algal polyphenols are promising and necessary to develop the basis for obtaining the most effective biologically active preparations for the pharmacological industry.

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